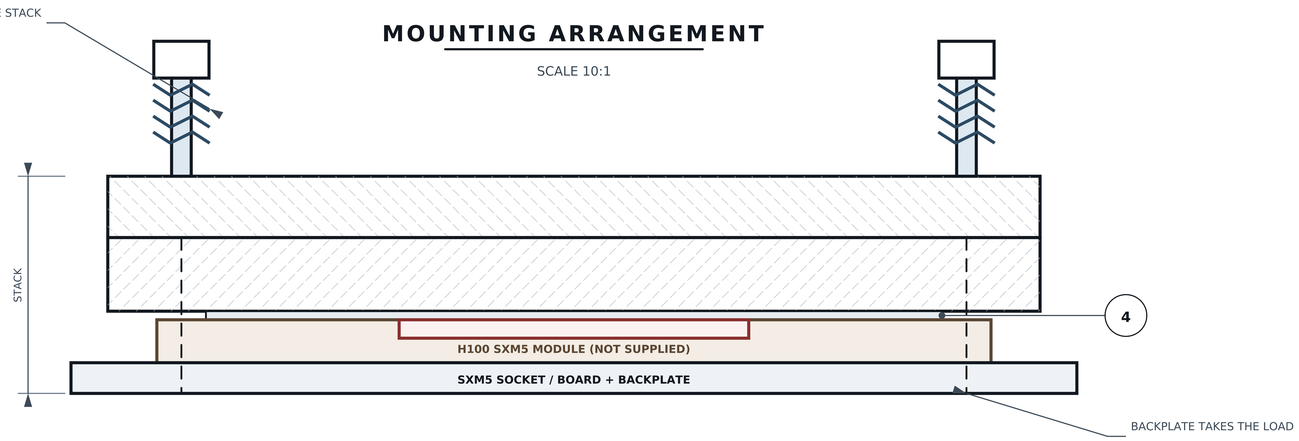
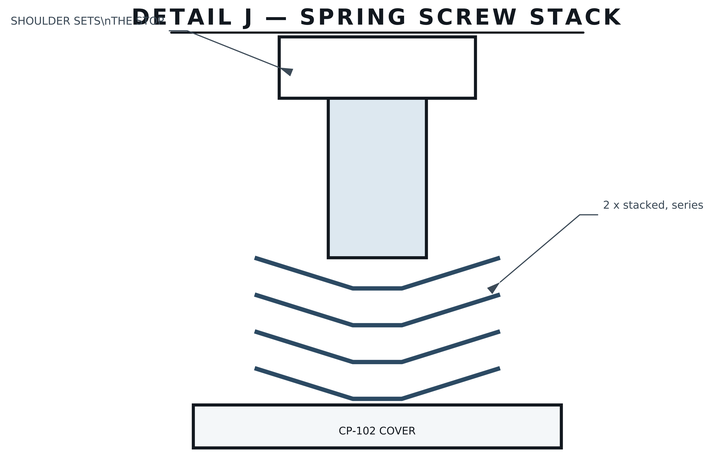
ASSUMED INTERFACE — CONFIRM EVERY DIMENSION AND THE PRELOAD AGAINST NVIDIA'S SXM5 MECHANICAL SPECIFICATION BEFORE MANUFACTURE

MOUNTING ARRANGEMENT

SCALE 10:1



DETAIL J — SPRING SCREW STACK



The shoulder, not the torque wrench, sets the preload.

Torque-only assembly on a bare die cracks silicon.

MOUNTING PARAMETERS

PARAMETER	VALUE	SOURCE
Total preload on the package	**TO BE TAKEN FROM NVIDIA SPEC**	**NOT SET HERE**
Screws	4 x M3 x 0.5, shouldered	this drawing
Spring element	2 x stacked, series	this drawing
Tightening	cross pattern, 3 passes to shoulder	this drawing
Cold plate flatness, die face	0.015 max	CP-101
Parallelism, plate to package	0.05 max	this drawing
TIM	PTM, k >= 13 W/mK, 60 um BLT	CP-104

WHY THIS SHEET IS NOT FINISHED

- An SXM5 module is a bare die on a socketed carrier. The socket, the load frame and the allowable preload are defined by NVIDIA, not by the cold plate designer.
- Too much preload cracks the die or the substrate. Too little starves the TIM bond line and the 4.0 K TIM budget on sheet 1 becomes 15 K or worse.
- Obtain the SXM5 mechanical and thermal design guide, then set the preload, the screw shoulder height and the Belleville stack to match it. Everything else on these six sheets is independent of that number.
- Do not substitute a torque specification for a shoulder stop. Torque on an M3 screw scatters by +/- 30 % with friction alone.

MOUNTING RISK REGISTER

RISK	MECHANISM	CONTROL
Die crack	**preload above the package rating, or uneven load**	**shoulder stop + cross tightening**
High TIM resistance	insufficient preload, or air in the bond line	preload to spec, controlled dispense
TIM pump-out	thermal cycling with a soft grease	use a phase-change material
Plate rocking	flatness or parallelism out of tolerance	0.015 / 0.05 per this sheet

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE: **SXM5 MOUNTING INTERFACE** — LOAD FRAME, PRELOAD AND TIM / DIMENSIONS ASSUMED — SEE BANNER	SCALE: **10:1** THIRD ANGLE PROJECTION	UNITS: **mm**	
DRAWING No.: **CP-200**	REV: **A**	SHEET: **5 / 6**	
DRAWN: AI-ASSISTED DRAFT	CHECKED: — PENDING —	MATERIAL: SEE BOM	FINISH: SEE NOTES	DATE: 2026-09-10